# Research Technology Core, College of Health Solutions

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## INTRODUCTION

The Research Technology Core (RTC) at the College of Health Solutions serves researchers in implementing their ideas using cutting edge technology to bring participants minimally invasive health support in the comfort of their own homes.

**Our solutions include:**

- Mobile health applications that support participants in monitoring health metrics, tracking behaviors, and receiving real-time feedback
- Wearable and IoT-integrated platforms that seamlessly connect devices like Fitbits, CPAP machines, and glucose monitors to deliver actionable insights.

By focusing on prevention, real-time feedback, and fostering healthy behaviors, we aim to improve health outcomes and advance the field of health research.

From ideation to deployment, we collaborate closely with researchers to understand their vision and create tailored solutions. Whether it's designing a mobile app, integrating wearable technology, or streamlining participant management, our team provides the technical expertise and innovation needed to bring research to life.

Together, we're not just building applications; we're shaping the future of health and improving lives through technology.

## AIMS

The RTC creates systems that enhance outcomes, streamline processes, and advance health research for researchers and participants alike.

**Enhanced Participant Engagement**:

- User-friendly designs and personalized feedback improve participant retention and make study participation accessible from home.

**Integration of Advanced and Scalable Technologies**:

- Integrates wearables, IoT, and mobile platforms for real-time feedback, healthier behaviors, and compliance with HIPAA and FDA standards.

**Support for Research Objectives**:

- Enables behavior change, longitudinal tracking, and tailored management solutions to bridge research and real-world applications.

The RTC advances health research by fostering engagement, enabling scalability, ensuring compliance, and transforming ideas into impactful applications.

## METHOD

Across multiple studies, an advanced suite of technological tools was deployed to enhance participant engagement, data collection, and study management. These innovative solutions created a scalable framework for health research, supporting diverse objectives. Key highlights include:

**Custom Mobile Application**:

- Served as a central platform for participant engagement, collecting biometric data, delivering real-time feedback, and providing visualizations to support adherence.

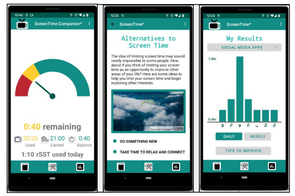
**Participant Management Web Application**:

- Facilitated study oversight, communication, and efficient data handling for researchers.

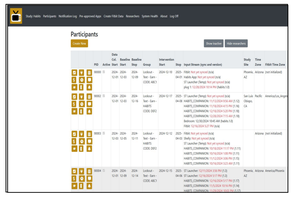
**Integrated Technologies**:

- Tracked and modulated screen time and device usage.

These integrated tools advance research by fostering engagement, enabling robust data collection, and streamlining management, demonstrating scalability across diverse health studies.



*Empowering Healthy Habits: A mobile app displaying personalized screen time usage, educational insights, and data visualizations to inspire informed choices and positive behavior change.*



*Streamlined Study Oversight: A participant management web application tracking app installations, Fitbit data, and Shelly plug communication for seamless participant engagement and data management.*

*Connected Health Monitoring: Seamless integration of Fitbit and Shelly Plug devices for comprehensive tracking of biometric data and device usage.*

## RESULTS

The development of mobile-integrated systems aims to empower health research with innovative, technology-driven solutions. These systems are designed to improve participant outcomes, streamline research processes, and advance scientific understanding.

**Enhance Participant Engagement and Retention**:

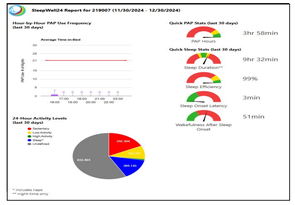
- User-friendly applications enable participants to engage in research, track progress, and receive feedback from home, fostering long-term retention and reliable data.

**Advance Health Behavior and Outcome Research**:

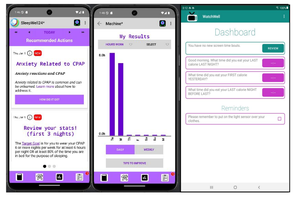
- Integrates advanced analytics, real-time monitoring, and longitudinal data tracking to provide insights that drive innovation and scientific discoveries; supporting studies of health behaviors, patterns, and interventions.

**Promote Behavior Change and Habit Formation**:

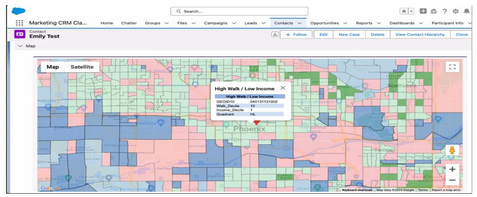
- Utilizes researcher determined education, motivational tools, feedback loops and reminders to foster healthier habits, driving long-term health improvements and behavior change insights.



*Empowering Better Care: A visualized sleep and CPAP usage report designed to keep participants and their providers informed and engaged in health improvement.*



*Transforming Health Research: Custom mobile applications showcasing curated education, sleep data visualizations, and dietary tracking for personalized, data-driven health insights.*



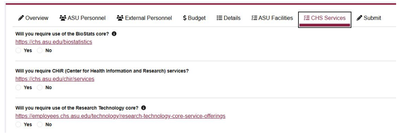
*Discover Your Path: An interactive neighborhood walkability app helping participants find enjoyable and accessible places to walk, promoting healthy lifestyles and community exploration.*

Mobile-integrated systems revolutionize health research by enhancing participant engagement, enabling scalable solutions, and driving scientific innovation. With these tools, researchers can bridge the gap between groundbreaking research ideas and real-world health improvements.

**Key Achievements to Date:**

- 10,000+ participants served
- 66 projects deployed
- 5 funded projects in pipeline

Request the RTC in the CHS Grant Management System!



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## CONCLUSIONS

The use of mobile applications integrated with wearable and IoT technologies in health research provides significant transformative advantages, enabling real-time data collection, personalized feedback, and improved participant engagement. These tools allow researchers to monitor behaviors and health metrics in natural settings, offering a more accurate representation of participant habits and outcomes. By reducing barriers to participation and fostering long-term adherence, these technologies deliver actionable insights into health behaviors, interventions, and outcomes.

## ACKNOWLEDGEMENTS

24h BEHAVIORS LABORATORY

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